Customer Experience Insights Dashboard

Project Brief Report | CSAT Analytics, October 2020

1. Introduction

Customer satisfaction is a core measure of service quality in a call-centre operation. This project analyses customer call data to understand call volume, communication channels, reasons for calling, response performance and customer sentiment, and turns the findings into recommendations for the company. The analysis and dashboard were built in **Power BI**.

2. Objective

- Collect and prepare the customer call data from the given database.
- Visualise call trends, channels, geography, calling reasons and response times.
- Analyse all features of the dataset and derive meaningful insights.
- Give the company practical suggestions to improve customer experience.

3. Dataset

The dataset holds **32,941 customer calls and 13 columns** from **October 2020**. Fields include customer ID and name, sentiment, CSAT score, call timestamp, day of the month, reason for calling, city, state, channel, response time, call duration and call centre. It covers 4 channels (Call-Center, Chatbot, Email, Web), 4 call centres (Los Angeles, Baltimore, Chicago, Denver) and 51 states.

4. Methodology

Data preparation (Power Query). The data was loaded from the project source and cleaned before analysis:

- The two date formats in `call_timestamp` (`MM/DD/YYYY` and `MM-DD-YYYY`) were standardised and converted to a date type.
- `Call_Day` was identified as the day of the month, so day-of-week columns were derived from the date.
- In `csat_score`, **20,670 records (63%)** held a placeholder 0 for customers who did not answer the survey. A `survey_status` column was created, and the zeros were replaced with null so they would not distort averages. Valid scores run from 1 to 10.
- Columns were renamed, data types corrected, and the data was checked for duplicates and blank values.

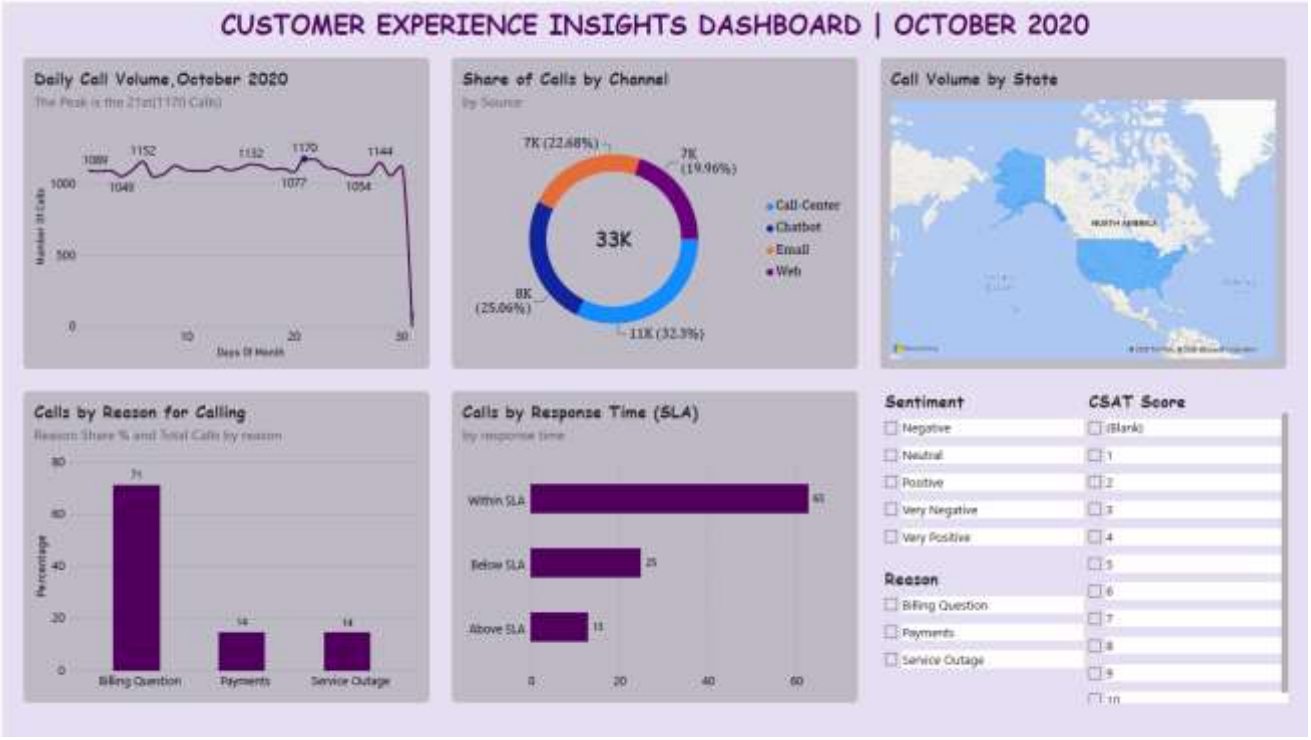
Measures (DAX). The report uses these measures: Total Calls, Surveyed Calls, Average CSAT Score, Average Call Duration, Total Call Hours, Survey Response Rate, Negative Sentiment %, Reason Share %, SLA Share % and Sentiment Calls with %.

5. Dashboard Overview

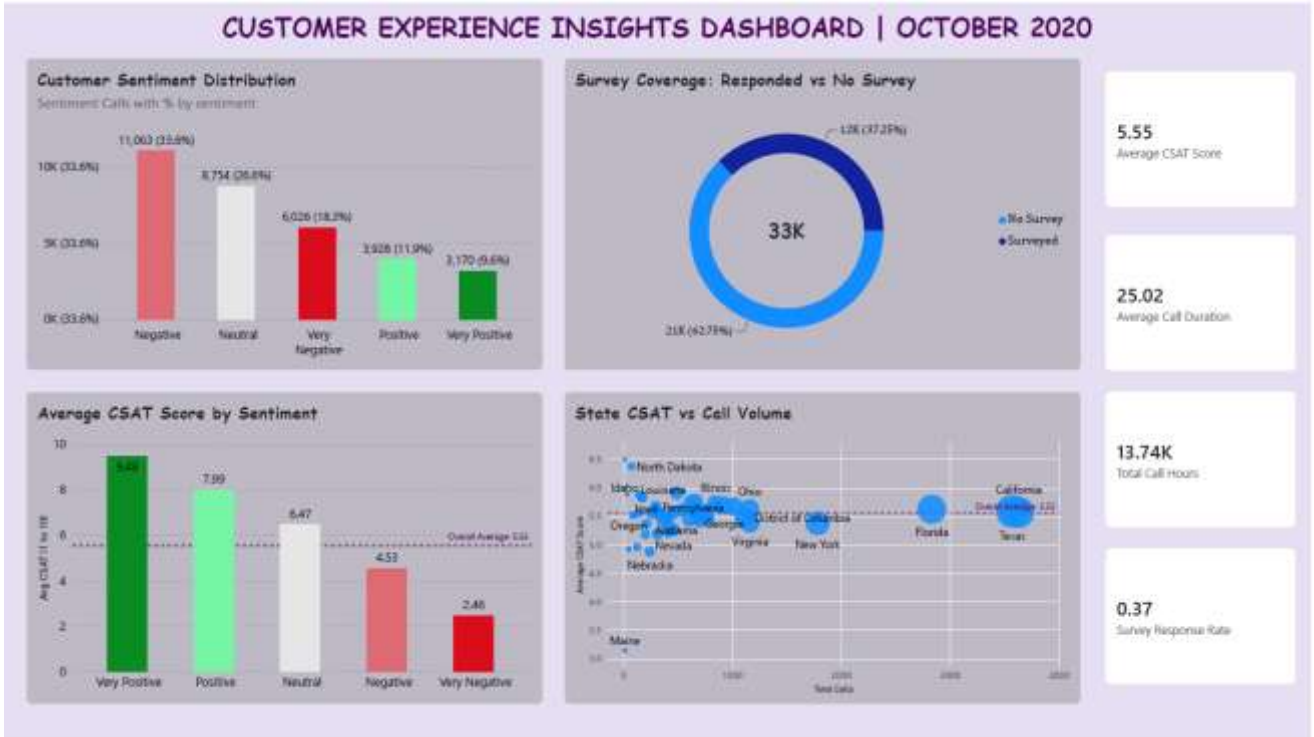
The report has two pages. **Page 1** covers volume and operations (trend, channel, geography, reason, response time) with slicers for sentiment, reason and CSAT score. **Page 2** covers customer experience (sentiment, survey coverage, state-level CSAT) with four KPI cards. The table lists every visual in the report; the snapshots follow.

Page	Visual	Type	Fields used
1	Daily Call Volume	Line chart	`Call_Day`, Total Calls
1	Share of Calls by Channel	Doughnut chart	`channel`, Total Calls
1	Call Volume by State	Filled map	`State`; tooltips: Average CSAT Score, Total Calls
1	Calls by Reason for Calling	Column chart	`reason`, Reason Share %
1	Calls by Response Time (SLA)	Bar chart	`response_time`, SLA Share %
1	Filters	3 slicers	`sentiment`, `reason`, `csat_score`
2	Customer Sentiment Distribution	Column chart	`sentiment`, Sentiment Calls with %
2	Survey Coverage	Doughnut chart	`survey_status`, Total Calls
2	Average CSAT Score by Sentiment	Column chart	`sentiment`, Average CSAT Score
2	State CSAT vs Call Volume	Scatter chart	`State`; X: Total Calls; Y: Average CSAT Score; size: Surveyed Calls
2	KPI cards	4 cards	Average CSAT Score, Average Call Duration, Total Call Hours, Survey Response Rate

Dashboard Snapshot: Page 1 (Volume and Operations)



Dashboard Snapshot: Page 2 (Customer Experience)



6. Key Insights

1. **Billing drives call volume.** Billing questions make up 71.2% of all calls, about five times Payments or Service Outage (14.4% each).
2. **Customer sentiment is poor.** About 52% of the 32,941 calls are Negative or Very Negative, consistently across channels, reasons and call centres.
3. **CSAT coverage is low.** Only 37.3% of calls have a survey score, with an average of **5.55 out of 10**. Counting the 63% placeholders as zeros would wrongly show 2.07.
4. **Channels satisfy equally.** Average CSAT is between 5.48 and 5.61 for Call-Center, Chatbot, Email and Web, so self-service channels match the call centre.
5. **Workload is concentrated.** Los Angeles (13,734 calls) and Baltimore (11,012) handle 75% of calls, while Denver handles about 8%.
6. **Volume is steady, with Friday and the 21st as peaks.** Daily volume stays near 1,100 calls; Friday is the busiest weekday (5,570 calls).
7. **Response time does not visibly change satisfaction.** 62.6% of calls are Within SLA, 24.7% Below SLA and 12.7% Above SLA, with similar CSAT in all three.
8. **Call duration has little effect.** The average call lasts 25 minutes, with almost no correlation to CSAT.
9. **CSAT follows sentiment.** Very Negative calls average 2.46 and Very Positive 9.49, so sentiment is a good guide to satisfaction even where no survey exists.
10. **State extremes are mostly small-sample noise.** Large states stay close to the 5.55 average, while states with few survey responses (for example Maine, 7 responses) show extreme scores. Rank states only where there are enough responses.

7. Recommendations

Priority	Finding	Recommendation	Expected impact
High	**Billing is 71.2% of calls.**	Simplify bills, send proactive bill explanations, and add a billing FAQ and chatbot flow.	Fewer billing calls, lower cost.
High	**About 52% of calls are negative.**	Run a root-cause review of negative billing calls, coach agents, and track negative sentiment weekly.	Lower negative share.
High	**Only 37.3% of calls have CSAT.**	Add a short post-call IVR or SMS survey and an in-chat rating prompt.	More representative CSAT.
Medium	**Chatbot and Web match the call centre on CSAT.**	Route routine billing and payment queries to Chatbot and Web.	Lower cost, same satisfaction.
Medium	**Los Angeles and Baltimore carry 75% of calls.**	Rebalance routing and plan extra staffing for Fridays and peak days.	Shorter queues.
Medium	**12.7% of calls are Above SLA.**	Confirm what the SLA labels mean, monitor the share and set a reduction target.	Lower service risk.
Low	**State CSAT varies on thin samples.**	Collect more survey responses in small states before acting on their scores.	Reliable state comparisons.

8. Limitations

The data covers one month (October 2020), 63% of calls have no CSAT score, and some states have very few survey responses, so state-level conclusions need caution.

9. Conclusion

Call volume is driven mainly by billing queries, and negative sentiment is high across the whole operation, which makes it a company-wide issue and not a problem of one channel or call centre. Acting on billing clarity, survey coverage and self-service adoption offers the greatest improvement in customer experience.

PREPARED BY:

Name:- ADARSH KUMAR PRASAD
Candidate Unique Number: CA04202689583